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Service Models in the Public Sector

RESEARCH

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ABSTRACT

This study investigates service models within the public sector and explores the government's role in delivering services. While extensive research has been conducted on business models in the private sector, the exploration of business models in the public sphere has received comparatively limited attention. To address this gap, our study adopted a comprehensive multi-method approach that combined both inductive and deductive methods. Through inductive analyses of real-world cases, we gathered rich and detailed information that allowed us to uncover patterns, trends, and the unique characteristics of these service models. Simultaneously, we drew upon the existing body of literature on service delivery, harnessing theoretical descriptions and concepts that guided our analysis and interpretation of the findings. Our work has led to the identification and comprehensive description of 45 different public sector service models. In closely analyzing the models, we were able to uncover five distinct roles that the government plays when operating these service models. This study offers new implications for the development and adaptation of innovative service models in the public sector.

HIGHLIGHTS

- We identify and analyze 45 service models in the public sector.
- We categorize five government roles in service models: Activator, Yield Shaper, Funder, Guardian of Public Goods, Performance Designer.
- Our study offers insights into adapting and innovating service models in the public sector.

ZUSAMMENFASSUNG

Diese Studie untersucht Servicemodelle im öffentlichen Sektor und beleuchtet die Rolle des Staates bei der Bereitstellung von Dienstleistungen. Während umfangreiche Forschungen zu Geschäftsmodellen im privaten Sektor durchgeführt wurden, hat die Erforschung von Geschäftsmodellen im öffentlichen Bereich vergleichsweise wenig Beachtung gefunden. Um diese Lücke zu schliessen, haben wir in unserer Studie einen umfassenden Multi-Methoden-Ansatz verwendet, der sowohl induktive als

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auch deduktive Methoden kombiniert. Durch induktive Analysen realer Fälle haben wir detaillierte Informationen gesammelt, die es uns ermöglichten, Muster, Trends und die einzigartigen Merkmale dieser Servicemodelle aufzudecken. Gleichzeitig haben wir auf das bestehende Wissen zur Servicebereitstellung zurückgegriffen und theoretische Beschreibungen und Konzepte genutzt, die unsere Analyse und Interpretation der Ergebnisse leiten. Unsere Arbeit identifiziert und beschreibt 45 verschiedenen Servicemodellen im öffentlichen Sektor. Bei der genauen Analyse der Modelle haben wir fünf unterschiedliche Rollen des Staates bei der Anwendung dieser Servicemodelle aufgedeckt. Die Studie liefert neue Implikationen für die Entwicklung und Anpassung innovativer Servicemodelle im öffentlichen Sektor.

RÉSUMÉ

Cette étude examine les modèles de service dans le secteur public et met en lumière le rôle du gouvernement dans la prestation de ces services. Alors que de nombreuses recherches ont été menées sur les modèles d'entreprise dans le secteur privé, l'exploration des modèles d'entreprise dans le domaine public a reçu relativement peu d'attention. Afin de combler cette lacune, nous avons utilisé une approche méthodologique complète combinant des méthodes inductives et déductives au sein de notre étude. Grâce à des analyses inductives de cas réels, nous avons collecté des informations détaillées qui nous ont permis de révéler les tendances, les schémas et les caractéristiques uniques de ces modèles de service. Parallèlement, nous nous sommes appuyés sur les connaissances existantes en matière de prestation de services et avons utilisé des descriptions et des concepts théoriques afin d'orienter notre analyse et notre interprétation des résultats. Notre travail identifie et décrit 45 modèles de service différents dans le secteur public. Une analyse approfondie de ces modèles nous a permis de mettre en évidence cinq rôles distincts du gouvernement dans l'application de ces modèles de service. Cette étude apporte de nouvelles implications pour le développement et l'adaptation de modèles de service innovants dans le secteur public.

1. INTRODUCTION

Analyzing service models in the public sector facilitates a more comprehensive understanding of public value creation and valuable insights into the intricate nature of public service delivery. Such an effort provides important knowledge regarding the complexities of public service delivery, shedding light on the roles of the government and stakeholders involved, and enables the development of increasingly effective strategies and measures to enhance service quality, efficiency, and public value. However, despite the importance of service models and innovation in the public sector (Cinar et al., 2024; Demircioglu, 2024; Demircioglu & Audretsch, 2024; Micheli et al., 2012; Wirtz et al., 2021), research in this area remains limited (Axelson et al., 2017). Service model thinking offers a systematic framework for evaluating and refining public service models (Wirtz et al., 2021). Yet, the general understanding of public sector service models remains incomplete, necessitating further research.

In contrast, the exploration of business models is deeply rooted in the private sector literature (Zott et al., 2011). Scholars have documented and described a variety of different models that explain how companies "do business" (i.e., how value is created, delivered, and captured) (Gassmann et al., 2014; Osterwalder & Pigneur, 2010; Zott et al., 2011). Business models are described as "stories that explain how enterprises work" (Magretta, 2002) and firms create value (Snihur & Wiklund, 2019; Zott & Amit, 2010).

Inspired by the private sector literature, public administration scholars have actively developed models and frameworks for exploring and understanding business models in the public domain. This ongoing endeavor stems from the recognition that public sector entities, like their private counterparts, engage in value creation, delivery, and capture. Martins et al. (2019, p. 49), for example, applied the Business Model Canvas approach and developed a Public Governance Canvas to serve as a tool for innovating public value creation in the public sector.

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However, the transference of business model concepts to the public sector is controversial due to misconceptions about public entities operating without business models (Schiuma & Lerro, 2017). Kaplan (2011, p. 6) challenged this notion, asserting that all organizations, irrespective of sector, are inherently involved in creating, delivering, and capturing value, and thus possess a business model. Notably, Bellone and Goerl (1992), Box (1999), Wirtz et al. (2021) and Lorenz, Heigl, Palmié and Oghazi (2024) provided valuable insights into the compatibility and challenges of adopting business-oriented approaches in government, as well as the need to balance managerial efficiency with democratic principles.

To address the nuanced yet well-recognized differences between the private and public sectors (Boyne, 2002; Hood, 1991), this study uses the term "service model" rather than business model. This term aligns more closely with the language used in public sector organizations and explicitly expresses the effort to develop and improve public service and deliver public value (Bloch & Bugge, 2013; Moore, 1995). Drawing on the existing literature (Agarwal et al., 2021; Gassmann et al., 2014; Osterwalder & Pigneur, 2010; Wirtz et al., 2021; Zott et al., 2011), we define service models in the public sector as government interventions to create, deliver, and capture public value. Analogous to business models in the private sector, service models in the public sector outline the strategies and mechanisms through which government entities fulfill their mission, address the needs of citizens, businesses, and other governmental agencies, and ensure the effective provision of public services.

Against this backdrop, the present research explores service models used by public sector organizations. Drawing upon the existing literature on business models (Gassmann et al., 2014; Wirtz et al., 2021) and acknowledging the paramount role of government in service models (Axelson et al., 2017), we investigate different service models and identify the government's role within each. Guided by this focus, we formulated the following research questions:

RQ1: What service models exist in the public sector?

RQ2: What role does government play in these service models?

To address these research questions, we opted for a mixed approach that combines both inductive and deductive methods (Mingers & Brocklesby, 1997). Our analysis yielded 45 service models operating in the public sector. They differ in how public services are delivered and value is created and captured. In each of the identified service models, the government was found to take a primary role. To identify the role of government within different service models, we analyzed each model based on the five key aspects of service models described by Schedler (2023): value creation, resource allocation, cost structure, revenue stream, and interactions with key stakeholders in the delivery of public services. Our analysis revealed five different government roles in service delivery: activator, yield shaper, funder, guardian of public goods, and performance designer. Our study contributes to the existing understanding of service models and the role of government in the public sector by providing a comprehensive analysis and description of the service models in operation in that context.

The remainder of this study is structured as follows. Section 2 provides background for the literature on business models. Section 3 outlines the research methodology, and Section 4 presents the research results. In Section 5, we discuss the findings and conclude with limitations and suggestions for future research.

2. BACKGROUND

2.1 BUSINESS MODEL APPROACHES IN THE PUBLIC SECTOR

Recent publications have demonstrated the importance of focusing on business models in the public sector, emphasizing the need for coordination among diverse public sector agencies, policymakers, and various stakeholders to achieve immediate and long-term value (Agarwal et al., 2021; Mol & Birkinshaw, 2009; Wirtz et al., 2021). Governments are increasingly under pressure to manage and foster innovation and enable change through innovative business models (Crosby et al., 2017; Martins et al., 2019). Especially when governments have limited resources, they need to find new ways to improve their services. For example, public institutions facing financial constraints can seek alternative funding mechanisms such as strategic partnerships with private entities (Blomqvist & Winblad, 2022). These partnerships may involve delegating

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project implementation to private companies in return for operational rights. However, the scarcity of resources is not only limited to financial resources but also includes human resources, IT infrastructure, and even input from customers (Agarwal et al., 2021; Chen & Perry, 2003).

Over the past decade, business models have emerged as a conceptual framework and research subject in both the public and private sector literature (Wirtz et al., 2021). However, the definitions and interpretations of the term "business model" and its purpose are diverse, numerous, and sometimes contradictory (Massa et al., 2017). This lack of consistency persists in both the private and public sector literature (Wirtz et al., 2021). Zott, Amit, and Massa (2011) have contributed to clarifying the concept by identifying key properties of business models, characterizing them as a new unit of analysis that operates at the system level, with a focus on activities and value creation (Zott et al., 2011). Others have adopted a broader definition, describing business models as depictions of organizations and their function in achieving goals such as profitability, growth, and social impact (Massa et al., 2017). This definition aligns with the properties outlined by Zott et al. (2011) and can be applied to the public sector due to its general nature.

There is a growing recognition that the public sector can drive value creation through business model innovation (Agarwal et al., 2021). After witnessing NPG and Digital Era Government, which emphasize holistic processes, multi-stakeholder collaboration, and shared value creation, business model thinking has become the next approach for public administration to improve public services and create public value (Feller et al., 2011; Micheli et al., 2012). Researchers are also increasingly applying tools developed for private sector business models to the public sector. For example, Martins et al. (2019) developed a Public Governance Canvas by applying the Business Model Canvas of (Osterwalder & Pigneur, 2010) to public sector organizations.

The study of public-sector business models is a prominent area of interest within the broader literature on service delivery although there remains a need for further theoretical and empirical exploration (Ranerup et al., 2016). Foss and Saebi (2017) noted a lack of comprehensive theoretical grounding and cumulative empirical inquiry in the current literature. Today, the precise understanding of business models in the public sector remains limited despite initial research efforts (De Vries et al., 2016). Therefore, there is a clear need to focus on defining business models, which we also refer to as "service models," and the role of government in the public sector context.

2.2 RESEARCH ON THE ROLE OF GOVERNMENT IN SERVICE MODELS

The government's role in service delivery has been repeatedly discussed in the literature (Boase, 2000; Borrás & Edler, 2020; Guenduez et al., 2023; Osborne et al., 2013; Osborne et al., 2016). A recurring theme revolves around whether the government should solely provide services or engage in shared provision with other actors. In ensuring the satisfaction of basic needs for the population, the government often steps in as a service provider or corrector of market and system failures that impede efficient and equitable service provision. Examples of services include education, healthcare, social security, infrastructure (such as roads, bridges, and water and electricity supply) and public transport. The government often takes the lead in delivering these services to ensure that all citizens have access to them, irrespective of their financial or other individual circumstances.

This level of government involvement leads to more control, responsibility, and centralization in service delivery, with government agencies and departments taking an active role. Education, for example, is a particularly critical policy area in which the government›s participation is substantial. Srivastava (2010) examined the government involvement in education provision in India, highlighting its roles as a *financier*, *manager*, and *regulator*. Similarly, in South Korea, where government has historically played a significant role in directing and guiding economic development, Cho (1992) identified the basic roles of the Korean Government as *subsidizer*, *regulator*, and *rule-setter*.

In contrast to this strong involvement of the government, alternative approaches also exist wherein the government collaborates and shares responsibility for service delivery with local and regional authorities and private sector actors (Bennett & Iossa, 2010; Carpintero & Petersen, 2016). In these cases, the role of the government becomes more supportive and facilitative. One such approach is the aforementioned NPM, which adopts market-oriented strategies that emphasize the government's role as a facilitator, grant autonomy to agencies, and promote

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private sector involvement in service delivery and outsourcing (Schedler & Proeller, 2000). By embracing market-oriented approaches like NPM, the government gains access to valuable resources and capabilities, including human resources, financial resources, and expertise. For example, in cases where governments face resource and capacity limitations, they may rely on public-private-partnerships (PPP) to leverage private sector resources and expertise and deliver public infrastructure projects and services (Pongsiri, 2002; Wang et al., 2019).

With the increasing digitization and advent of new technologies like AI and big data (Guenduez et al., 2020), the role of government is once again under scrutiny. Borrás and Edler (2020) provided a comprehensive perspective by examining the roles of the government in transformative socio-technical systems, with a focus on illustrative cases such as smart cities, cryptocurrencies, and automated vehicles. Through their analysis, they identified 13 distinct roles of the government, including *observer*, *facilitator*, *initiator*, and *watchdog*, finding that different modes of governance exhibit unique combinations of these roles. Taking a more specific focus on AI, Guenduez and Mettler (2023) investigated the government's role in this emerging technology. They identified four key roles that the government can play in the context of AI: *regulator* (reducing risks and threats, reconciling AI with human values); key *enabler* (investing in infrastructure and training, creating a secure legal framework); *leader* (building and promoting AI capabilities through partnerships with the government/industry and universities); and *user* (enhancing public sector efficiency, supporting decision-making processes using AI). These studies by Guenduez and Mettler (2023) and Borrás and Edler (2020) demonstrate that the government›s role in navigating the complexities of emerging technologies, urban developments, and significant societal challenges is multifaceted and versatile.

3. METHOD

To answer our research questions, we opted for a sequential multi-method approach (Mingers & Brocklesby, 1997) consisting of five distinct phases. These phases are outlined in detail in Table 1 and elaborated upon below. Steps 1 to 3 were specifically designed to address Research Question 1 whereas Step 4 was dedicated to answering the second research question.

Table 1 Methodical Approach.

STEP	OBJECTIVE	EXAMPLE
Deductive analysis	Use the Business Model Navigator, which includes service models, as a reference point and analyze the applicability of these models in the public sector.	"Hidden revenue" eliminated, "Crowdfunding" included on the list of service models in public administrations.
Inductive analysis	Identify additional real-life public sector service models by exploring the existing literature and examining real-world examples.	"Taxpayers' money" included as additional service model in the public sector.
Expert discussions	Evaluate service models through discussions with practitioners and academics to identify weaknesses and drive improvements.	Edit wording in "Outsourcing" service model.
Clustering	Conduct a comparative analysis of service models to identify, describe, and cluster the roles of government in public service delivery then report back and discuss them with the expert group.	Government in its role as "activator."

In the first phase of our research, our primary objective was to compile a comprehensive list and provide descriptions of existing service models in the public sector, thus contributing to the investigation of our first research question. We started by utilizing the widely recognized Business Model Navigator developed by Gassmann, Frankenberger, and Choudury (2020) as a foundational framework for our investigation. At its core, the Business Model Navigator presents and analyzes diverse business models (see Table SI A.1.), offering a comprehensive overview of how private sector organizations engage in value creation, delivery, and capture. While the Navigator is primarily focused on the private sector, we have taken its principles as a starting point to identify analogous service models in the public sector.

To conduct a literature search, we employed a two-pronged search strategy utilizing the Web of Science (WoS) for its curated social science focus and Google Scholar database for its broad academic coverage (Harzing & Alakangas, 2016). We applied search terms such as "government,"

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"public services," "public sector," "government services," "public management," and "public administration." We identified these terms by reviewing the search terms used in literature reviews on the public sector and public administration in general (e.g., Gomes de Sousa et al., 2019; Zuiderwijk et al., 2021) and public service delivery (e.g., Tummers et al., 2015) and business models in the public sector (e.g., Wirtz et al., 2021) in particular. In our search, we also incorporated business models mentioned in the Business Model Navigator, such as "crowdfunding" (Gassmann et al., 2020). Our goal was to determine if the business models identified from the existing literature were also present in the public sector. Our analysis uncovered a range of established service models within the public sector, including "service platform" "orchestration," and "payment by results." By contrast, we did not find any publications dealing with some of the business models common in the private sector, such as "cash machine," "open business," and "supermarket" (see, Table SI A.1; Gassmann et al., 2020; Gassmann et al., 2014). We thus excluded them from further analysis. At this point, we were able to adopt 24 of the service models from the private-sector typology of Gassmann et al. (2014) even if they sometimes deviated from the private-sector models in their application in the public context.

In the second step, we pursued an inductive search to identify novel service models exclusive to the public sector and thus not listed in the Business Model Navigator. To specifically explore service models within the public sector, we utilized the WoS and Google Scholar databases for our literature search. This search, conducted from April to November 2023, included the terms "service models," "business models," "public sector," "public service," and "government service." Boolean operators like "AND" and "OR" were employed to refine our search. Additionally, we used the Google search engine with the same search terms to find real cases of applied service models. We entered keywords like "public sector service models" and "government service delivery" and then refined our search by including terms such as "case study" and "real-world examples." This approach led to the discovery of 19 new service models, such as the "taxpayers' money" model. This step resulted in a preliminary list of 43 service models.

In the third step, we engaged in discussions with a diverse panel of experts who possessed in-depth knowledge and expertise in the field of public service delivery. Using convenience sampling (Flick 2013), we assembled a panel consisting of four professionals from the government sector responsible for service development, four consultants from the public sector, and three academics. All participants were part of an existing group of experts for public administration in Switzerland whom we anticipated would bring a diverse range of experiences and understandings of service delivery in the public sector. Our goal was to gather insights and feedback on the list of service models we had compiled. For this reason, we conducted focus groups and expert discussions, drawing upon the benefits of soliciting feedback simultaneously from multiple experts (Nyumba et al., 2018). Through these discussions, which included both moderated and non-structured requests for feedback, we sought to refine and enhance our understanding of the service models.

Furthermore, we organized a half-day workshop with these experts (Thoring et al., 2020). For the workshop, we prepared individual "playing cards" representing each service model. In the Appendix (see Supporting Information, Figure SI B.1), we provide two illustrative examples of our Service Model Cards. During the workshop, participants engaged in an exercise of randomly selecting service models by drawing cards from the set. They then discussed how their assigned task could be accomplished using the chosen service model. Alternatively, some participants used a rapid-fire approach, presenting and brainstorming ideas for each model in a brisk manner. In these workshops, we also discussed the role of the government in each service model. We kept these discussions open in order to gain as many insights as possible. Based on these discussions, we clustered the service models according to the primary roles of the government. This dynamic process allowed us to continuously enhance the models by refining their descriptions and making adjustments; eventually, we introduced two new service models. As a result, we finalized a comprehensive list of 45 public sector service models.

In this final step, our central focus shifted to answering the second research question, or identifying the role of the government in the service models. Our analysis focused on understanding the role of the government in value creation, resource allocation, the structure of costs, revenue streams, and interactions between the government and key stakeholders, including citizens, businesses, and civil society organizations. Building upon these detailed role descriptions, we proceeded with clustering to identify commonalities and patterns across the various service delivery models. This enabled us to categorize the government's role in public

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service delivery into five distinct categories offering a comprehensive representation of the diverse ways in which the government provides services. We reported the results back to the expert group and held further discussion about the clustering. We then completed the analysis when no further suggestions for improvement were received.

4. RESULTS

Our analysis revealed 45 different service models. A detailed description of each, along with two specific examples, can be found in the Supporting Information. In the following, we present our research findings on service models in the public sector with regards to the five identified roles of government: activator, yield shaper, funder, guardian of public goods, and performance designer. Despite some degree of overlap between individual service models in terms of collaboration with other actors, the use of market mechanisms, and incentives, the five roles illustrate the various ways in which the government engages in service delivery. This spectrum ranges from actively providing services, as exemplified by the performance designer, to roles such as the activator, which centers on creating incentives, activating other actors, and fostering collaboration with external partners. Furthermore, these roles exhibit variations in how services are financed; for instance, some rely on public funding, exemplified by the funder, while others explore alternative revenue streams, as seen in the case of the yield shaper.

Figure 1 provides a brief overview of these roles, which we discuss in more detail in the following sections.

Activator: The state activates others to become involved in the fulfillment of a public task.

Yield Shaper: The government organizes the service model around the ways in which it can generate income from performing a public function.

Funder: The government promotes the accomplishment of a public task by using public (government) funds awarded through a budgeting process.

Guardian of Public Goods: By law, the state has certain public goods (e.g., water rights), the use of which it can transfer to private parties.

Performance Designer: The state fulfills a public task by providing certain services. The design of these services can vary depending on the service model.

Figure 1 Roles of the government in service models.

4.1 ACTIVATOR

The government plays a critical role as an activator by activating, enabling, and empowering others to participate in public service delivery, and, in some cases, delegating these tasks entirely to third parties. This approach, rooted in the concept of the guarantee state (Schedler & Proeller, 2000), entails a combination of strategies employed by the government. As an activator, the government mandates that service providers deliver public services directly to users. The "contracting out" service model is commonly used, wherein the government outsources some or all of the public tasks to reduce its direct involvement. Examples include preschool education, waste disposal, healthcare, and welfare services (Vining & Globerman, 1999). Through outsourcing, the government aims to enhance service quality, improve efficiency, and reduce overall expenditures (Blomqvist & Winblad, 2022) while also bypassing bureaucratic barriers and enabling the government to focus on its core tasks (Prager, 1994).

As an activator, the government also embraces a participatory approach by involving external actors in value creation processes and knowledge generation, fostering collaboration and inclusivity. For instance, the government may adopt the developer platform service model by providing some of its data to offer insights into its performance and activities (Janssen & Estevez, 2013). This encourages collaboration, transparency, and innovation in the co-production of public services. Additionally, the government activates civil society by providing open government data and modules, promoting an open dialogue and facilitating transparency, and ultimately enabling the co-production of better public services (Bonina & Eaton, 2020). A notable example is the Ushahidi platform launched by the government of Kenya in 2008,

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which allows the crowd-sourcing of violence reports through various channels and supports the creation of custom applications by its users (Janssen & Estevez, 2013). A government can provide digital modules on a platform, share ownership of the infrastructure with private parties, and form partnerships to leverage resources and expertise. Network effects gained in this way between service providers and platform users promote engagement.

In summary, the government's role as an activator is to empower and involve actors in public service delivery. Through strategies like contracting out and initiatives such as the developer platform model and open government data, this role fosters collaboration, transparency, and innovation. By providing digital modules and forming partnerships, the government promotes engagement, ultimately facilitating effective service delivery and empowering stakeholders to create public value. The service models where the government acts as an Activator are summarized in Table 2.

Table 2 Activator Service Models.

SERVICE MODEL	MODE OF OPERATION	LITERATURE EXAMPLE & ILLUSTRATIVE CASES
Contracting Out	The government mandates that a service provider deliver a specific public service directly to users.	Literature example: • Blomqvist and Winblad (2022) Illustrative cases: • NYC Plaza Program • Street lighting in Bulgaria
Crowdsourcing	The government involves a large number of external actors in value creation, innovation processes, or the generation of knowledge.	Literature example: • Liu (2017) Illustrative cases: • Australian Parliament open data offering • Bogotà's participatory online crowdsourcing platform
Developer Platform	The government or a third party provides digital modules on a platform that other communities can use in developing their own applications.	Literature example: • Janssen and Estevez (2013) Illustrative cases: • The multi-governmental initiative GovStack • eGovernment platform Mobile Age of the European Union
Fractional Ownership	The government shares ownership of an infrastructure that generates public value with individual private parties who may also be users of the public service.	Literature example: Goedkoop and Devine-Wright (2016) Illustrative cases: • Shared ownership of solar panels in England • Middelgrunden wind turbines in Denmark
Government as a Platform	Creation of network effects among both the providers and buyers of services. This creates a motivation to use the platform.	Literature example: Jung et al. (2019) Illustrative cases: • NHS Jobs platform in the UK • eCitizen platform in Kenya
Orchestrator	The government coordinates the interaction of different groups of actors in such a way that public value is created.	Literature example: Shaw et al. (2019) Illustrative cases: • Sharing City Seoul project in South Korea • Organ transplant system in Spain
Self-Service	The government lets citizens perform certain public tasks themselves.	Literature example: Paulin (2013) Illustrative cases: • Green sponsorships in Bonn • Neighborhood patrol program in Phoenix

4.2 YIELD SHAPER

In this role, the government combines market-based principles with government involvement to deliver public services while also creating funding opportunities. While acting in such a role, the government recognizes that financing activities through taxpayers' money alone is not the only option and actively explores alternative sources of funding.

One way the government leverages market dynamics is through the auction service model. The government harnesses the willingness of private buyers to pay by conducting auctions for special public services and prestigious objects (Chow, 2020; Woo et al., 2008). Since 1973, the Hong Kong

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government has been holding open auctions for license plates where motorists can bid for the license plate they want. Numbered plates made during the Lunar Year 1990 achieved an average price of over $400,000 USD, which was paid to the Hong Kong government, thus highlighting the success of this service model (Woo et al., 2008). Another illustrative example of a service model in which the government acts as a yield shaper is the user pays or polluter pays principle. This service model demands that individuals or entities responsible for causing certain (often external) costs to society should be held accountable and required to pay a levy to offset those costs (Luppi et al., 2012). The user or polluter pays service model serves two purposes; first, it establishes private financing for public tasks, and second, it creates an incentive to minimize the generation of societal costs. For instance, the government takes responsibility for environmental damage, recovering costs from the original polluter through subrogation (Luppi et al., 2012).

Our analysis shows that various modes of government exist in the literature on the provision of public services. Besides auctioning special services to the highest bidder (auction) and charging fees for the use of services (user pays), these also include financing service creation through community contributions (crowdfunding), levying fixed taxes on services (flat rate), generating services from customer data (leverage customer data), collaborative revenue sharing (revenue sharing), selling exclusive network advantages (sell your network), providing taxpayer-funded services (taxpayers' money), and generating income from waste (trash-to-cash). Each mode represents a unique approach to financing and delivering public services, highlighting the diverse strategies that governments employ as a yield shaper. The service models for the Yield Shaper role are summarized in Table 3.

Table 3 Yield Shaper Service Models.

SERVICE MODEL	MODE OF OPERATION	LITERATURE EXAMPLE & ILLUSTRATIVE CASES
Auction	Special public services are sold at auction to the highest bidder. The model leverages customers' willingness to pay for prestige objects.	Literature example: Chow (2020) Illustrative cases: • License plate auction in Hongkong • Auction platform for old cars in Australia
Crowdfunding	The creation of a service is financed by contributions from the community of potential users. The crowd can organize itself as a private company with ownership participation, but the model can also take more informal forms.	Literature example: Lee et al. (2016) Illustrative cases: • Pedestrian bridge in Rotterdam • Renewable energy investments in Scotland
Flat Rate	The government levies a tax on a service, but this is defined as fixed irrespective of the quantity purchased.	Literature example: Ehtasham (2022) Illustrative cases: • Refuse collection fees in Singapore • Passes for swimming pools in Canada
Leverage Customer Data	The government generates data from a service that it then makes available for other services or to the public.	Literature example: Kim et al. (2014) Illustrative cases: • Provision of social and economic data for business startups in Mexico • COVID-19 payments in Denmark
Revenue Sharing	Collaborative creation of a public service leads to revenue that is shared among the different actors.	Literature example: Wang and Liu (2015) Illustrative cases: • Free public Wi-Fi in Kansas City • Expansion of the superhighway in Pakistan
Sell Your Network	The government is part of a larger (exclusive) network and sells this advantage.	Literature example: Carrera (2014) Illustrative cases: • Golden passport scheme in Malta • E-Residency in Estonia
Taxpayers' Money	The government provides a public service, which is financed by taxpayers' money. This is free of charge for the beneficiary.	Literature example: Mules (1998) Illustrative cases: • Urban parks in Beijing City • Primary education in Mexico

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SERVICE MODEL	MODE OF OPERATION	LITERATURE EXAMPLE & ILLUSTRATIVE CASES
Trash-to-Cash	The government provides itself with a source of income from the waste of its citizens or from its own waste.	Literature example: Kabirifar et al. (2020) Illustrative cases: • Waste heat and energy in Berlin • Recycling from road construction in Australia
User pays	The government charges a fee for extraordinary burdens caused by an actor.	Literature example: Luppi et al. (2012) Illustrative cases: • Motorway tolls in Europe • Polluter pays principle

4.3 FUNDER

The government's role as a funder includes various strategies to support and promote public services and initiatives. In the role of funder, the government focuses its efforts on enabling the fulfillment of public tasks through financial benefits. At the forefront is the understanding that private parties need financial incentives to take on a public task. One way the government supports private parties is by facilitating access to capital. This is done through loan guarantees that allow private companies to raise capital at a lower cost. By guaranteeing some or all of the loan amount, the government encourages private companies to take on tasks that would otherwise be financially challenging.

An illustrative example of a service model in which the government acts as a funder is the provision of subsidies for private social institutions in a city. It is becoming increasingly common for social entrepreneurs and non-profit organizations to provide social services to citizens. The government can support such private initiatives with financial contributions from the government budget. In this way, it ensures that the public task is fulfilled without having to complete the task itself. In Canada, for example, there is a long tradition of government funding of nonprofit organizations. Among the beneficiaries is the Canadian Red Cross in Toronto, which thanks to governmental subsidies is able to directly provide valuable services to seniors, refugees, and victims of disaster (Akingbola, 2003).

However, the government can also take the opposite approach and waive or reduce taxes or fees instead of actively providing financial support. By waiving or reducing financial obligations, the government encourages private companies to engage in activities that align with its goals, thereby promoting the development of initiatives that benefit society. For instance, incentive schemes for renewable energy in Indonesia include both an income tax reduction on energy development projects and an import duty exemption for machinery needed for the construction of power plants (Damuri & Atje, 2012).

Overall, the government plays a multifaceted role as a funder, using a range of strategies to fund public initiatives and ensure their success and sustainability. Besides the aforementioned examples, the government might also provide assurances for loans, enabling private parties to raise capital at low cost (guarantees) and offer payment-by-results contracts whose compensation is tied directly to the impact of the services provided (performance-based contracting). To foster self-sustainability, the government could also allocate start-up funding, offering start-up loans free of charge contingent on the service's success (seed money). Moreover, individuals may also benefit from vouchers that facilitate access to necessary services (voucher). These funding mechanisms play crucial roles in facilitating public initiatives and ensuring the delivery of essential services. The service models where the government acts as a Funder are summarized in Table 4.

Table 4 Funder Service Models.

SERVICE MODEL	MODE OF OPERATION	LITERATURE EXAMPLE & ILLUSTRATIVE CASES
Guarantees	The government enables the private party to raise capital at a low cost by guaranteeing the full loan or a portion of the loan.	Literature example: Bradshaw (2002) Illustrative cases: • COVID-19 credit guarantee scheme in Ireland • Small business loan guarantees in California
Payment by Results	The government pays the service provider based on the impact the service has.	Literature example: Janus and Keijzer (2015) Illustrative cases: • Energy performance contracts in Boston • Results-based aid

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SERVICE MODEL	MODE OF OPERATION	LITERATURE EXAMPLE & ILLUSTRATIVE CASES
Seed Money	The government provides start-up funding to a public service that subsequently becomes self-sustaining.	Literature example: Bakos (1997) Illustrative cases: • Revolving water fund in Madhya Pradesh • Care support points in Bavaria
Start-Up Loans	The government provides a private party with a (free) loan to set up a service, which must be repaid if the service is successful.	Literature example: Kratzer and Kato (2013) Illustrative cases: • Empowerment loans in Tanzania • Solar power production in Sri Lanka
Subsidies	The government supports the private fulfillment of tasks with cash benefits. These can be investment or operating contributions.	Literature example: Zhang et al. (2022) Illustrative cases: • Euro-tickets in Germany • Nationwide internet connection in Ireland
Vouchers	The government supports persons by contributions to the consumption of necessary services.	Literature example: Bahta and Musara (2022) Illustrative cases: • Food vouchers during COVID-19 in South Africa • Intercantonal contributions for free choice of study location in Switzerland
Waivers	The government promotes a specific task by waiving or reducing taxes or fees.	Literature example: Damuri and Atje (2012) Illustrative cases: • Housing and development public rentals scheme in Singapore • Renewable energy tax incentives in Indonesia

4.4 GUARDIAN OF PUBLIC GOODS

As a guardian of public goods, the government assumes the role of governing and overseeing various critical resources. This role is based on the recognition that certain resources such as water, virgin forests, clean air, and even data are of great importance, scarce, and require thoughtful regulation, governance, and protection. Consequently, the government plays a critical role in defining and regulating the appropriate use of these goods, ensuring compliance with established rules and regulations.

One service model that illustrates the role of the state as a guardian of public goods is the granting of licenses. Licenses are legal permission that allow actors to use intellectual property for commercial purposes. By regulating the licensing process, the government ensures that intellectual property rights are protected. With this service model, the government protects innovation and promotes economic growth. In order to pursue this aim and at the same time increase transparency and accountability, various governments, including the UK, make government data available through a public license framework (Lee, 2016).

Maintaining registries that serve as a central database is another service model that highlights the state's role as a guardian of public goods. Registries enable the identification and verification of a wide range of information relating to identities, rights, and obligations. Through this service, governments preserve important information and provide clarity and transparency on legal issues. Therefore, in an attempt to enforce their regulations, the governments of Niger and Swaziland register small and micro-enterprises through a logit framework (McPherson & Liedholm, 1996).

The government's multifaceted role as a guardian of public goods also encompasses granting the private sector rights to extract natural resources and charge third parties for service costs (concessions), holding the authority to grant or deny permits for activities that have societal impacts (permits), overseeing access to networks and competition in sectors characterized by natural monopolies (regulator), and employing sanctions to penalize behaviors that violate legal standards (sanctions). All of these service models enable effective governance, ensuring appropriate resource management, activity regulation, and enforcement of legal compliance. Table 5 highlights the service models associated with the government's role as a Guardian of Public Goods.

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SERVICE MODEL	MODE OF OPERATION	LITERATURE EXAMPLE & ILLUSTRATIVE CASES
Concessions	The government grants the private sector the right to extract a natural resource and charge third parties for the cost of producing the service.	Literature example: Örnberg and Ólafsdóttir (2008) Illustrative cases: • Alcohol sales monopoly in Iceland • Mineral extraction in India
Permits	The government grants or denies permits for activities that involve emissions for society.	Literature example: Ono (2002) Illustrative cases: • Work permits in Bangladesh • Street music in Dublin, Ireland
Licenses	A license allows an actor to use a licensor's intellectual property for commercial purposes.	Literature example: Lee (2016) Illustrative cases: • Open government licenses in Canada • UK data service in England
Registrator	The government runs a register that clarifies relations, rights, and duties.	Literature example: Yadav et al. (2022) Illustrative cases: • Digital land register in Hong Kong • Trade register entries in the Isle of Man
Regulator	The government regulates access to a network and competition in an area characterized by natural monopolies.	Literature example: Moran (2002) Illustrative cases: • Regulatory Offiice for Network Industries in Slovakia • Railway safety regulators in South Africa
Sanctions	The government punishes behavior that does not comply with the law.	Literature example: Zahnow et al. (2022) Illustrative cases: • Speeding fines in Switzerland • Architecture for reintergration in the Netherlands

Table 5 Guardian of Public Goods Service Models.

4.5 PERFORMANCE DESIGNER

In the role of the performance designer, the government's main focus is on designing services that meet the specific needs and preferences of beneficiaries. This includes considerations such as user experience, resource allocation, and the overall effectiveness of service delivery. By designing and customizing services, the government aims to achieve the best outcomes for the public. The role of the government extends beyond simply designing the service themselves; it also involves defining the methods and processes of value creation. In this way, the government strives to ensure that services are optimized to meet the specific needs of the beneficiaries.

One such service model is the add-on, in which the government finances the basic service and provides additional services for a fee. This model allows users to customize their experience and improve the overall performance of the service. Another service model that aligns with this one is mass personalization. This model allows users to personalize standardized service modules to create customized solutions tailored to the specific needs of citizens or businesses. These include, for example, the pre-filled tax return system in Malaysia, which reduces cost and complexity while increasing data security and privacy (Ibrahim, 2013).

Our analysis revealed many other service models that reflect the government's role as a performance designer. Whether the government grants a paid, temporary right of use for part of its property (rent instead of buy); exchanges goods, services, or rights with third parties (barter); provides analogous services in digital form (digitization); offers different quality levels of a service and sells them at different prices (label the level); or makes data or source code for its applications available so that third parties can further refine and improve them (open source); the underlying goal remains the same: develop services that maximize efficiency and effectiveness, meet beneficiary needs and requirements, and create value for the general public. The service models where the government acts as a Performance Designer are summarized in Table 6.

Table 6 Performance Designer Service Models.

SERVICE MODEL	MODE OF OPERATION	LITERATURE EXAMPLE & ILLUSTRATIVE CASES
Rent Instead of Buy	The government grants a paid, temporary right of use for part of its property. The user is effectively a tenant but has extensive rights of use.	Literature example: Hoppe and Thimm (2018) Illustrative cases: • Bike rental system in Constance • Community use of schools initiative in Ontario
Add-on	The basic service is financed by the government, but additional services can be used for a fee.	Literature example: Gilson et al. (1995) Illustrative cases: • Extracurricular support through "Club Engage" in South Africa • Snow removal on private property in Solothurn
Barter	The government exchanges goods, services, or rights with third parties without money being involved.	Literature example: Waters and Williams (2011) Illustrative cases: • Twitter labels for government account • Allmend corporation in Switzerland
Digitization	The government provides (formerly analogous) services in digital form.	Literature example: Lindgren et al. (2019) Illustrative cases: • Digitization plan in New Zealand • Digital payment systems in Liberia
Experience Selling	The government provides emotional value to a "natural" offering through clever labeling.	Literature example: Buurma (2001) Illustrative cases: • Roger Federer promotes Corona measures in Switzerland • Highway sponsorship in the US
Label the Level	The government offers different quality levels of a service and sells them at different prices with easily understandable names.	Literature example: Cutler (2002) Illustrative cases: • Fees for the post office in Tanzania • Train travel in France
Lock-In	Technical measures or penalties are used to keep switching costs high so that users do not switch providers.	Literature example: Vink et al. (2019) Illustrative cases: • Transfer between universities in Switzerland • Loss of citizenship in the Netherlands
Mass Personalization	Personal configuration of standard modules creates individualized solutions for individual citizens or companies.	Literature example: Ibrahim (2013) Illustrative cases: • Pre-filled tax documents in India • MyGovNL in the Netherlands
Object Self-Service	Objects order materials autonomously.	Literature example: Ayele et al. (2020) Illustrative cases: • Waste collection in Barcelona • Proactive bridge monitoring in Norway
Open Source	The government provides data or source code for its applications so that third parties can further refine and improve them.	Literature example: Kalampokis et al. (2011) Illustrative cases: • Chicago data portal in the US • Digital testbed framework in Estonia
Outsourcing	The government is responsible for performance but purchases input from suppliers.	Literature example: Tahir and Wang (2019) Illustrative cases: • EduApp4Syria in Norway • Intelligencec agencies using AWS in the US
Sell Your Services	Government sells services in a commercial way to third parties (public or private customers).	Literature example: (Guenduez et al., 2023) Illustrative cases: • KSD Schaffhausen in Switzerland • GovCloud in Denmark

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SERVICE MODEL	MODE OF OPERATION	LITERATURE EXAMPLE & ILLUSTRATIVE CASES
Sensor as a Service	The government supplies sensors to the private sector to provide services to the public.	Literature example: García et al. (2020) Illustrative cases: • Irrigation system in Barcelona • Parking in Tržič in Slovenia
Shop-in-Shop	In addition to the core offering, the government (or a leased-in actor) offers other services in the same infrastructure.	Literature example: Askim et al. (2011) Illustrative cases: • One-stop stores in Australia • Post coffee in Fujian, China
Solution Performance Designer	The government organizes on the principle of "one face to the customer," combined with a comprehensive service package in specific situations.	Literature example: Ladychenko and Tunitska (2019) Illustrative cases: • "Single window" for import of goods in Ukraine • Point of single contact in Austria
Standby	The government ensures that a particular service is available to beneficiaries when needed.	Literature example: Perry and Lindell (2003) Illustrative cases: • The HyperSport responder in Dubai • Medical care at –70°C in Antarctica

5. DISCUSSION AND CONCLUSION

In this study, we set out to address two research questions. First, we examined service models operating in the public sector, with the goal of gaining a comprehensive understanding of their characteristics. Second, we explored the role of the government within these service models. To answer these questions, we employed a sequential multi-method approach (Mingers & Brocklesby, 1997), enabling us to conduct an in-depth analysis. We identified 45 service models in the public sector. We categorized these models into five roles of government based on the degree of government involvement. The result was a comprehensive typology of service models currently operating in the public sector. In the following, we discuss the contributions of our study and conclude with limitations and future research opportunities.

5.1 CONTRIBUTIONS TO THE LITERATURE

The role of government in service delivery has evolved through various epochs, each characterized by distinct theoretical frameworks and emphases (Dickinson, 2016; Hood, 1991; Osborne, 2006; Schedler & Proeller, 2000; Torfing & Triantafillou, 2013). In this comprehensive investigation, we undertook an extensive analysis of both the literature and international practices, unveiling the intricate landscape of government activity in service delivery. Our primary objective was to meticulously identify and categorize diverse service models that encapsulate the multifaceted roles governments play. This research resulted in the identification of 45 distinct service models, each offering illuminating perspectives on the myriad governmental functions in public service delivery and insights into the government's various roles.

Within the context of these roles, we observed striking commonalities across diverse functions. Whether donning the hat of an activator, yield shaper, funder, guardian of public goods, or performance designer, the government demonstrates a commitment to fostering innovation, mobilizing external resources, and exploring alternative funding avenues. In all these roles, the overarching goal remains steadfast: to maximize the efficiency and effectiveness of services, as achieved through strategic collaborations with external actors and the judicious application of market-based principles.

In conclusion, our study offers a significant contribution to understanding the manifold roles governments play in service delivery. These insights not only build upon existing theoretical frameworks like NPM and NPG but also provide a comprehensive understanding of the diverse functions performed by governments. This enhanced comprehension can serve as a valuable foundation for policymakers and practitioners seeking to optimize service delivery and enhance the well-being of citizens.

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5.2 CONTRIBUTIONS TO PRACTICE

Our study has several implications for practice. First, the Service Model Typology, consisting of 45 models, is highly relevant for practitioners in the public sector, including planners, procurement experts, and commissioners involved in service model redesign. It serves as a valuable tool for decision makers seeking to select appropriate service models based on specific needs and goals. By providing a comprehensive overview of service models and roles for governments to use in enhancing public service delivery, the typology addresses a crucial gap in the literature. It equips practitioners with the guidance needed to develop innovative solutions for task performance in specific situations. Planners can leverage the typology's findings to design service delivery models that incorporate collaboration, innovation, financing mechanisms, resource governance, and service design, resulting in more effective services. Procurement experts can now explore alternative funding approaches, identify new opportunities, and establish partnerships for sustainable service delivery. Commissioners can utilize the typology to design governance structures, promote public value creation, ensure compliance, and customize services to enhance overall quality and effectiveness. By leveraging these insights, practitioners drive positive changes, benefiting the government and citizens alike.

Second, when developing new services, a key challenge is identifying the crucial factors that enable its success, including relevant actors, benefits, and funding (Poister, 2010; Poister & Streib, 2005). In a manner similar to that of the private sector (Gassmann et al., 2020), service models can serve as valuable heuristics to assist in identifying these factors. They help answer critical questions such as: Who are the beneficiaries? What value do we deliver? How is that value created? How do we finance it? Moreover, these questions prompt government representatives to reflect on their role in service delivery, encouraging critical examination of the government's current role and fostering exploration of alternative roles that can enhance service delivery.

Third, beyond the role of government, a review of the entire service model can help foster innovation in administrative practices. When redesigning an existing service model, it is essential to generate creative ideas that facilitate the achievement of public functions in new and inventive ways. By questioning established service models and drawing insights from other models, administrative workers can stimulate the development of innovative service models and explore novel approaches to service delivery.

5.3 LIMITATIONS AND FUTURE RESEARCH

This study has certain limitations that highlight potential avenues for future research. First, while we identify 45 service models, we caution against interpreting this as an exhaustive typology. Given the dynamic nature of public sector organizations, additional models are likely to emerge over time. The exploratory design of this study also suggests that different datasets or administrative contexts could yield alternative sets of models. Using the framework proposed by Gassmann et al. (2014) and providing comprehensive examples, we establish a foundation that future research can build upon to identify similar models in various public administration settings. Future studies could expand upon this work by analyzing a broader spectrum of datasets across diverse administrative contexts, potentially uncovering region- or government-specific service models. Moreover, longitudinal studies could prove valuable for tracking the development and evolution of these models over time, offering insights into how public sector organizations adapt their service strategies in response to shifting needs and challenges.

Second, while Web of Science (WoS), Google Scholar, and Google Search Engine provided a strong foundation for our search, recognizing the inherent limitations of any single database (Harzing & Alakangas, 2016), we acknowledge that incorporating additional resources could potentially yield further literature and uncover new insights that could expand our list of service models.

Third, this study primarily focuses on simple service models. In practice, more complex arrangements integrating multiple models exist (Hodge & Greve, 2019; Klasen, 2022). Public services can be improved by combining different service models. In this respect, service provision strategies (i.e., whether a specific service model is adopted or various service models are combined) may vary depending on the objective and specific requirements. In this study, we focused on identifying and describing individual service models and did not explore potential combinations. To improve waste management, for example, governments can employ a toolbox approach. This might involve contracting out waste collection to private companies (Contracting Out) while also offering

residents a self-service recycling option (Self-Service). Furthermore, incentivizing waste collection companies based on the volume of recycled waste (Payment by Results) could be implemented. This combined approach aims to boost efficiency, empower citizens, and incentivize environmentally friendly practices. Similarly, for improving public transportation, a government agency could leverage public knowledge through crowdsourcing for app features (Crowdsourcing), then create a developer platform (Developer Platform) with public transport data and tools. This "Government as a Platform" approach fosters innovation and creates valuable citizen services. Finally, to support after-school programs for low-income children, the government can act as an Orchestrator, connecting non-profits offering programs with funding and the families that need them. Tools like Seed Money (grants to develop programs), Vouchers (subsidies for participation), and Orchestration can stimulate program development, increase accessibility, and create a coordinated support system. Further research is needed to analyze the different service delivery strategies (i.e., whether a single service model or combination of different models is chosen) and the rationale guiding those decisions. Furthermore, the various service models are closely interwoven with the diverse roles of government. A combination of different service delivery models also implies those multiple roles. Future research that focuses on exploring combinations of service models will also enrich our understanding of the different roles of government in service delivery.

In conclusion, despite the aforementioned limitations, this study provides valuable insights into a number of service delivery models and the role of government in public service delivery. By illuminating these aspects, this study opens new avenues for research that can improve government practice and promote service innovation.

ADDITIONAL FILE

The additional file for this article can be found as follows:

- **Supporting Information.** Supplementary data – Service Models in the Public Sector. DOI: https://doi.org/10.5334/ssas.222.s1

COMPETING INTERESTS

The authors have no competing interests to declare.

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